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15 April 1976
SUPERSEDING
MIL-E-1/1717(NAVY)
31 May 1972

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

1/ TYPE 8906

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, planar, ceramic and metal

See figure 1

Mounting position: Any

Weight: 1.8 ounces (48 grams) nominal

ABSOLUTE RATINGS:

Parameter: / Unit:	F GHz	Ef V (Note 1)	Eb Vdc	epy kv	Ec Vdc	ib a (Note 2)	Ib mAdc	ic a	tp μs	Du ---	Pp W
Maximum:											
Anode pulsed osc or amp:	3.0	6.0 ±5%	---	4.5	-150	5.0	---	2.5	6.0	0.0033	10
Grid pulsed osc or amp:	3.0	6.0 ±5%	3,500	---	-150	5.0	---	2.5	6.0	0.0033	10
TEST CONDITIONS:	---	6.0	600	---	Adj	---	25	---	---	---	---

ABSOLUTE RATINGS:

Parameter: / Unit	Pg W (Note 3)	tk sec (min)	TE °C (Note 4)	T (anode shank) °C (Note 4)	Barometric pressure, reduced mmHg (Note 6)	Cooling --- (Note 5)
Maximum:						
Anode pulsed osc or amp:	2.0	60	250	250	---	Conduction and convection
Grid pulsed osc or amp:	2.0	60	250	250	35	Conduction and convection
TEST CONDITIONS:	---	300	---	---	---	Note 7

GENERAL:

Qualification - Required (see note 18)

1/ Replaces EIMAC Y-572

Ⓐ denotes changes

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METHOD	REQUIREMENT OR TEST	TES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1	17		0.65	II	R	50	---	Meg
1211	Insulation of electrodes	-	Eb = Ek = 0; Ec = -500 Vdc	0.65	II	Ec	-4.5	-9.5	Vdc
1261	Electrode voltage (1) (grid)	-		0.65	II	Ic	---	-2.0	μAdc
1266	Total grid current	-		0.65	II	etd	---	200	v
1231	Pulsing emission	-	eb = ec = etd/is = 10.0 a; tp = 3 μs (max); prr = 600 (max)	0.65	II	If	0.90	1.05	A
1301	Heater current	-							
	Quality conformance inspection, part 2			---	---	---	---	---	---
---	Resonance test	9, 10	No voltages applied	---	---	Ec	---	-30	Vdc
1261	Electrode voltage (2) (grid)	-	Eb = 1,000 Vdc; Ec/Ib = 1.0 mAdc	---	---	Cin	7.0	9.0	pF
						Cgp	1.85	2.10	pF
1331	Direct-interelectrode capacitance	9	No voltages applied; fixture per Drawing 158-JAN	---	---	Cout	---	0.06	pF
				---	---	Po	6.0	---	W (useful)
1236	Power oscillation (pulse)	9, 12	F = 3.0 GHz (min); epy = 3.5 kv; Ec = -1.5 Vdc (min); Rg/Ib = 12.0 mAdc (max); Ic = 7.0 mAdc (max); Ef = 5.8 V						
---	Power gain	13	F = 1,100 - 100 MHz; Ebb = 2,200 Vdc; Ecc = -50 Vdc; tp = 3 μs (min); Du = 0.002 (min); pd = 400 w	---	---	po	1.8	---	kw
	Quality conformance inspection, part 3			---	---	---	---	---	---
(A) ---	Life test	16	Group C; Ef = 6.0 V; filament standby; t = 500 hours	---	---	Δib	---	25	%
				---	---	ΔCgp	---	0.15	pF
---	Life-test end point	16		---	---	Ep	---	250	mVac
---	Variable-frequency vibration	9, 11, 14	F = 55 to 500 to 55 Hz; Accel = 10 G peak (min); Ebb = 300 Vdc; Rp = 10,000 ohms; Ec/Ib = 10 mAdc	---	---	---	---	---	---
---	Torque	9, 11, 15	No voltages applied	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	[illegible]	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
1042	Shock, specified pulse	11	Condition A; no voltages applied; fixture per Drawing 280-JAN	---	---	---	---	---	---
---	Torque and shock-test end point:								
1266	Total grid current	-		---	---	Ic	---	-10	μAdc
1002	Barometric pressure, reduced	8, 11	Pressure = 35 mmHg (max); voltage = 2,000 Vac; TA = 30° ±10° C	---	---	---	---	---	---

NOTES:

1. The transit-time heating effect of the cathode shall be compensated for by a reduction in heater voltage after dynamic operation of the tube has started. The back heating is a function of frequency, grid current, grid bias, anode current, duty cycle, and circuit design and adjustment. There is an optimum heater voltage which will maintain the cathode at the correct operating temperature for a particular set of operating conditions. A maximum variation of ±5 percent from optimum is permitted. No reduction in heater voltage is required up to and including 500 MHz.

2. The regulation, or series anode supply impedance, or both, shall limit the instantaneous peak current, with the tube considered as a short circuit, to a maximum of 10 times the specified maximum current rating.

3. The maximum instantaneous peak grid voltage for grid-pulse conditions shall be within the range of +250 to -750 volts.

4. Sufficient conduction and convection cooling shall be provided to limit the envelope and anode shank temperatures to the specified maximum value under all operating conditions. Reliability will be seriously impaired if this maximum is exceeded. Where emphasis is placed on long and reliable life, lower temperatures should be maintained.

5. Sufficient cooling shall be provided for all seals to limit temperature in accordance with note 4. In cases where long life and consistent performance are factors, cooling in excess of minimum requirements is normally beneficial.

6. Operation at this altitude is possible in a suitably designed circuit.

7. In all electrical tests involving application of heater voltage, sufficient conduction, convection, or forced-air cooling may be provided for all seals to limit temperatures.

8. Voltage shall be 60 Hz ac applied between anode and grid. No other voltages shall be applied. There shall be no evidence of failure as indicated by arc-over. This is not a destructive test.

9. Other tube contact configurations may be used provided the tube contact area remains unchanged and the socket, jig, or cavity gives equal performance. Mounting of the socket, jig, or cavity may be at the option of the manufacturer.

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NOTES: -Continued

10. Grid-anode resonance. Test in cavity in accordance with Drawing 278-JAN. Cavity shall resonate at 1,354 ±2 MHz with tuning slug in accordance with Drawing 277-JAN at TA = 25° ±5° C.

    Grid-cathode resonance. Test in cavity in accordance with Drawing 283-JAN. Cavity shall resonate at 1,719 ±2 MHz with tuning slug in accordance with Drawing 277-JAN at TA = 25° ±5° C.

    When plotted on graphs of resonant frequency versus grid-anode capacitance and resonant frequency versus grid-cathode (input) capacitance, the tube under test shall be represented by a point within a parallelogram whose four corners are located by the following points:

Points	Capacitance (pF)		Frequency (MHz)	
	Cgp	Cgk	Fgp	Fgk
1	1.85	7.0	2010	1760
2	1.85	7.0	2040	1800
3	2.10	9.0	1940	1720
4	2.10	9.0	1970	1760

(A) 11. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted. None of the testing shall be considered as destructive except in case of failure.

12. The applied voltage pulse shape shall be measured with a noninductive resistor of 1,150 ohms ±2 percent inserted in place of the tube. The pulse shape shall be: tp = 3.0 μs ±10 percent, tr = 0.4 μs maximum, and tf = 0.7 μs maximum. The pulse repetition rate (prr) shall be adjusted so that Du = 0.0025 ±5 percent with the above measured pulse length. Test in cavity in accordance with Drawing 279-JAN. The cavity shall be connected to a load with a VSWR less than 1.5/1. The oscillator output coupling and the grid or cathode resistor may be adjusted for maximum power output.

13. Test shall be conducted in power amplifier cavity as shown on figure 2, or equivalent. Driving power is defined as the net power delivered to the amplifier cavity input terminals and the reflected power shall be subtracted from the incident power to obtain the net driving power. The output tuning shall be adjusted for maximum power output.

14. The tube shall be mounted in the socket in accordance with Drawing 276-JAN and vibrated with simple harmonic motion. The peak acceleration over the frequency range shall be within ±20 percent of the reference acceleration at 100 Hz. The frequency shall vary from 55 to 500 Hz and return to 55 Hz with approximate logarithmic progression and shall require 4 minutes minimum, 6 minutes maximum, to traverse the range. Each tube shall be vibrated for 30 minutes in each position X and Z except that if the cumulative result of test on 50 or more tubes of a construction show that more than 75 percent of the tubes have a higher output voltage in one position, subsequent measurements need to be taken only in the position giving the higher readings. The voltages specified herein shall be applied to the tube during vibration. The value of the alternating voltage, Ep, produced across the resistor, Rp, as a result of vibration, shall be measured with a suitable device. This device shall have an appropriate voltage range, and shall have the ability to measure, with an error less than 10 percent, the rms value of a sine wave of voltage at all frequencies from 20 to 20,000 Hz. The value of the vibrational output, Ep, shall not exceed the limit specified herein at any point in the sweep frequency range during the last complete cycle of cycling vibration.

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NOTES: -Continued

15. The torque test shall be performed as follows:

    (a) The tube shall be held securely at the cathode connection. A force of 5 pounds shall be applied to the heater cup without perceptible shock. This test may be made by applying the force at right angles to the inside of the cup at a point 7/64 (2.77 mm) ± 1/64 (0.41 mm) inch from the cathode end of the tube. An approved equivalent method may be used. The heater cup shall not loosen or short circuit to the cathode connection.

    (b) A rotational torque of 15 inch-pounds shall be applied between anode and cathode without shock.

    (c) A rotational torque of 40 inch-pounds shall be applied between anode and grid without shock.

    (d) A rotational torque of 30 inch-pounds shall be applied, both clockwise and counter-clockwise, between the anode cooler and the anode contact surface, with no resulting loosening of the anode cooler or damage to the tube.

(A) 16. At zero hours, measure the grid-anode capacitance (Cgp), and establish the drive conditions necessary to obtain 3.0 amperes peak anode current with an anode voltage of 1,000 Vdc and a bias voltage of -40 Vdc. The pulse width of the modulator shall be 2 μs minimum and the duty cycle shall be 0.025 maximum.

    At the completion of life test, Cgp is measured and the change from the original value (ΔCgp) may not exceed the specified limit; under the same operating conditions and with the drive level determined at zero hours, check the anode current at the end of life, and the change from the original value (Δib) may not exceed the specified limit.

17. All tests listed under quality conformance inspection, part 1, shall be performed at the conclusion of the holding period.

18. In addition to the requirements specified herein, compatibility testing shall be required in the AN/ARN-52(V) Tacan Set. A sample of 20 tubes shall be supplied to NAD Crane, Code QEEL on a noninterference basis. Of the 20 tubes to be supplied, a sample of 4 tubes shall be selected for life tests. Each tube shall achieve a minimum of 500 hours operation. In the event of no failures, the sample shall be considered as acceptable when a total of 1,600 hours have been accumulated.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

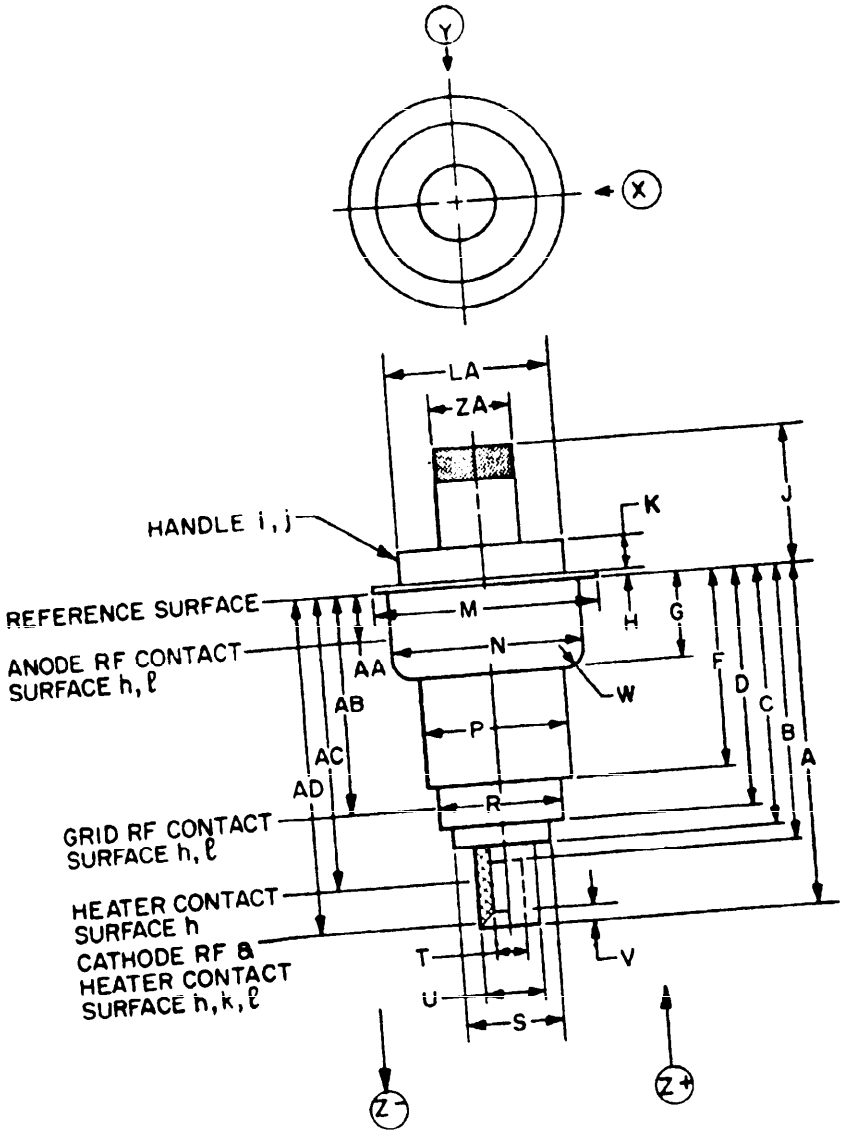
User activities:
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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NOTE: When type 8906 tubes being tested to this specification sheet have been made in the same production run as the JAN-7815 tubes and the listed dimensions are identical, then the one sample may represent both types.

FIGURE 1. Outline drawing of electron tube type 8906.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 2			
A	1.815(46.10)	1.875(47.62)	
B		1.534(38.96)	
C		1.475(37.46)	
D	1.289(32.74)	1.329(33.76)	
F	.970(24.64)	1.010(25.65)	
G	.462(11.73)	.477(12.12)	
J	.766(19.46)	.826(20.98)	
N	1.025(26.04)	1.035(26.29)	e, m
R	.655(16.64)	.665(16.89)	e, m
T	.213 (5.41)	.223 (5.66)	f, m
U	.315 (8.00)	.325 (8.26)	e, f, m
Quality conformance inspection, part 3			g
H		.040 (1.02)	
K		.185 (4.79)	
M	1.180(29.97)	1.195(30.35)	
P	.752(19.10)	.792(20.12)	
S		.545(13.84)	
V		.086 (2.18)	
W		.100 (2.54)	
LA	.840(21.34)	.860(21.84)	
ZA	.427(10.85)	.447(11.35)	
Electrode contact areas			n
AA	.035 (.89)	.361 (9.17)	a, e
AB	1.185(30.10)	1.265(32.13)	b, e
AC	1.534(38.96)	1.728(43.89)	c, f
AD	1.475(37.46)	1.815(46.10)	d, e, f

FIGURE 1. Outline drawing of electron tube type 8906 - Continued.

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NOTES:

a. Anode rf contact surface and reference dimension for eccentricity measurements.
b. Grid rf contact surface and reference dimension for eccentricity measurements.
c. Heater contact surface and reference dimensions for eccentricity measurements.
d. Heater and cathode rf contact surface and reference dimensions for eccentricity measurements.
e. The total indicated runout of the anode and grid contact surface with respect to the cathode contact surface shall not exceed .020 (.51 mm).
f. The total indicated runout of the cathode contact surface with respect to the heater contact surface shall not exceed .012 (.30 mm).
Ⓐ g. These dimensions shall be tested monthly when in continuous production, with sampling as follows:

n1 = 4	c1 = 0	where c2 represents the total allowable failures for the first and second samples combined.
n2 = 4	c2 = 1;	

In case of failure after double sampling, the failing dimension(s) shall become quality conformance inspection, part 2, AQL 6.5, inspection level S3. After three consecutive successful submissions the sampling may revert to the double sampling plan.

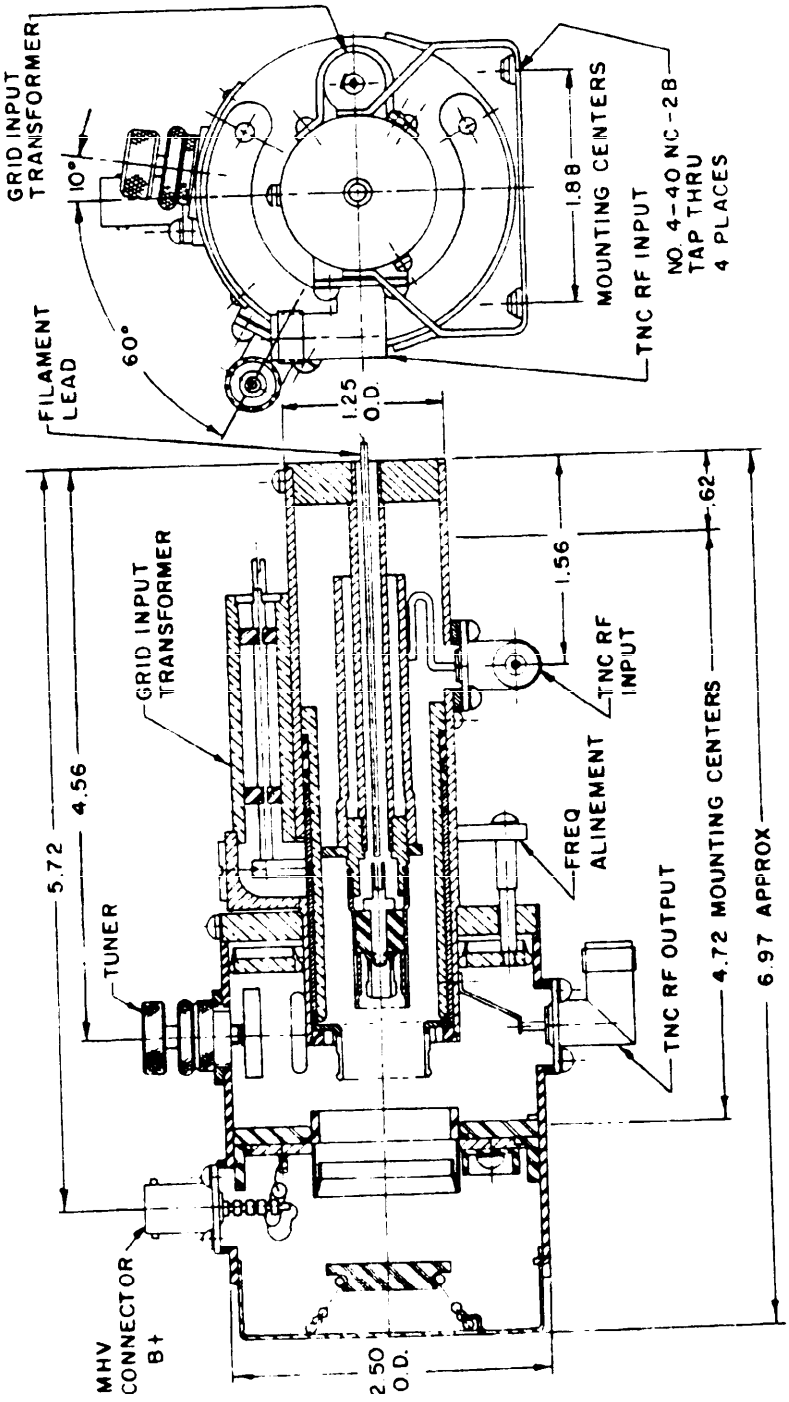
Ⓐ h. Silver plated 30 MSI minimum.
i. Plating not required over radiator or handle support of copper, aluminum, or approved equivalent.
j. This surface shall be used for measurement of anode shank temperature.
Ⓐ k. Inner edge of heater and outer edge of cathode rf connection shall be free from burrs and sharp edges. Insulated material between heater and heater-cathode shall be securely affixed.
l. Total indicated runout (T.I.R.) of contact surfaces shall be gaged from center line of reference and shall be as follows. Note g shall apply.

Contact surface	T.I.R. (Maximum)	Reference
Anode	0.020	Cathode
Grid	0.020	Cathode
Heater	0.012	Cathode

m. Diameters N, R, T and U shall apply throughout entire contact areas as defined by dimensions AA, AB, AC, and AD respectively.
n. Dimensions in electrode contact areas table are for socket design purposes and are not intended for inspection purposes.

FIGURE 1. Outline drawing of electron tube type 8906 - Continued.

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INCHES	MM
.62	15.75
1.25	31.75
1.56	39.62
1.88	47.75
2.50	63.50
4.56	115.82
4.72	119.89
5.72	145.29
6.97	177.04

NOTE: All dimensions are for reference only.

FIGURE 2. Pulse amplifier cavity.

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